

Conformal Prediction as an Averaged Limit: notes on the corrector under Markov dependence

Peter Cotton
`peter.cotton@microprediction.com`

August 2026, substantially revised September 2026

Abstract

These are technical notes, not a contributions paper. Pooling calibration residuals averages a fast variable over its invariant law, which is the averaged limit of homogenization, so split conformal prediction is correct to leading order exactly when the fast variable is fast relative to the forecast horizon. The gap it leaves is a corrector, and for a Markov chain that corrector solves a Poisson equation whose solution also generates the calibration variance. On GARCH at the persistence the finance literature fits, the marginal guarantee holds exactly while conditional coverage runs from 98% to 74% across volatility deciles, which is the corrector made visible. We then record four attempts to turn this into a procedure, all of which failed, and what is already in the literature. Section 11 is the most useful part of the document.

1 Status of this document

What follows is a record rather than a proposal. The mathematics in Sections 3 to 8 is correct and checked numerically to machine precision. The reading in Section 9 is, we think, the useful contribution, and it is a way of seeing rather than a theorem. Everything that was once offered here as a procedure has been withdrawn, and Section 11 says why.

Three things are known and we claim none of them. Effective sample size for autocorrelated data is Bayley and Hammersley (1946), in the title. The mean return time to a set is Kac's lemma. And the calibration-conditional coverage of a conformal threshold under stationary mixing, with the long-run indicator variance, clustered calibration and an effective sample size, is Ramos et al. (2026). That paper covers the ground a reader might expect these notes to claim, and it does so properly.

What may remain is narrow: the exact Poisson representation of the realized history-conditional coverage in Theorem 1, and the observation that the same solution family supplies the calibration variance. We have not searched the Markov chain literature for the identity itself, and it should be assumed to exist until someone does.

2 The exact object

Take a stationary time series of nonconformity scores and the split-conformal threshold, an order statistic of a calibration block. Two things are true at once and are usually conflated. The calibration scores are dependent, so the order statistic used as the threshold is not the one an i.i.d. analysis assumes. And the test point is generated by a dynamical process whose conditional

law given the current state need not equal the invariant mixture. The exact object of study is neither the invariant law nor the marginal coverage. It is the realized history-conditional coverage of a random order-statistic threshold, and both effects act on it.

It is tempting to say that ordinary conformal prediction “calibrates the invariant law.” Under dependence that is not an exact guarantee. Dependence alters the law of the order statistic, and the conditional law given the current state alters the law of the future test score. Our contribution is that one solution family of a Poisson equation governs both perturbations: the transient gap between the current-state and invariant score laws, and the long-run calibration noise of the dependent empirical quantile. The identity is exact and asymptotics-free, and we verify every algebraic claim numerically to machine precision.

Contribution and related work. Two effects of this kind have been described before, and we treat them as background. Transported-beta analyses separate a test-side coverage transport from a deformation of the calibration order statistic (Ramos et al., 2026). Distributional conformal prediction for Markov processes constructs a conditional probability integral transform from estimated transition CDFs (Dai et al., 2026), and concentration-based penalties quantify coverage loss under non-exchangeable and Markovian data (Oliveira et al., 2024; Zheng and Proutiere, 2024). Our narrow claim is this: we give a Poisson representation of realized history-conditional coverage and use the same solution family to obtain both the transient state correction and a finite-sample martingale-based tolerance correction for dependent calibration data. The nonparametric price of a fixed-point guarantee is the subject of a companion note (Cotton, 2026), whose local modulus reappears in Section 13.

3 The exact coverage decomposition

Let (W_t) be a Markov chain with transition kernel P and invariant distribution π , on a state space rich enough that all emission randomness is absorbed, so the score is a deterministic readout $R_t = r(W_t)$. This convention matters below: with the emission randomness inside W_t , the calibration-variance calculation needs no separate emission martingale. For each threshold q write the coverage function and the invariant score CDF

$$H_q(w) = \mathbf{1}\{r(w) \leq q\}, \quad F(q) = \pi H_q. \quad (1)$$

Assumption 1 (Ergodic chain with bounded Poisson solutions). The chain is uniformly ergodic with invariant law π , and there is a jointly measurable map $(w, q) \mapsto \chi_q(w)$ solving the Poisson equation

$$(I - P)\chi_q = H_q - F(q), \quad \pi \chi_q = 0, \quad (2)$$

with $\sup_q \|\chi_q\|_\infty < \infty$ over the quantile range used. For a finite irreducible aperiodic chain both hold automatically, and the reader may keep that case in mind.

Let $R_{(k)}$ be the k th order statistic of the calibration scores $R_1, \dots, R_n$, let $\mathcal{F}_n$ be the calibration history, and let $\mu_n(\cdot) = \mathbb{P}(W_n \in \cdot | \mathcal{F}_n)$ be the filtering distribution of the current state. If the test score is observed ℓ steps after the calibration block, its realized coverage is

$$C_{n,\ell,k} = \mathbb{P}\{R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n\} = \mu_n(P^\ell H_{R_{(k)}}), \quad (3)$$

the ℓ -step transported coverage function averaged against the filter. The invariant law is the wrong reference by exactly the solved Poisson equation. Since $F(q)$ is constant and P^ℓ fixes constants,

$$P^\ell H_q - F(q) = P^\ell (H_q - F(q)) = (P^\ell - P^{\ell+1})\chi_q. \quad (4)$$

The first equality already writes the state discrepancy without solving anything. What the Poisson solution χ_q adds, and the reason we carry it, is that the *same* χ_q generates the additive-functional martingale and its variance in Section 6.

Theorem 1 (Exact conformal coverage corrector). *Under Assumption 1, for every calibration threshold $R_{(k)}$,*

$$C_{n,\ell,k} = F(R_{(k)}) + \mu_n[(P^\ell - P^{\ell+1})\chi_{R_{(k)}}]. \quad (5)$$

The first term is the coverage of the random calibration threshold under the invariant score law. The second is the exact current-state correction. If the state is observed at time n , so W_n is $\mathcal{F}_n$ -measurable, the filter is a point mass and

$$C_{n,\ell,k} = F(R_{(k)}) + (P^\ell - P^{\ell+1})\chi_{R_{(k)}}(W_n). \quad (6)$$

No asymptotics enter. Given the Poisson solution, (5) is an algebraic identity holding for every n, ℓ, k . A direct simulation on a finite chain confirms both sides of (6) agree to machine precision (Section 16).

4 A transported order-statistic representation

Suppose F is continuous and set $U_i = F(R_i)$. Each U_i is marginally uniform under stationarity, though the sequence need not be independent, and monotonicity of F gives $F(R_{(k)}) = U_{(k)}$. Define the state-dependent coverage transport

$$\tau_{\ell,w}(u) = P^\ell H_{F^{-1}(u)}(w) = u + (P^\ell - P^{\ell+1})\chi_{F^{-1}(u)}(w). \quad (7)$$

Then (6) reads $C_{n,\ell,k} = \tau_{\ell,W_n}(U_{(k)})$, which separates the problem cleanly. The law of $U_{(k)}$ carries the calibration dependence. The map τ_{ℓ,W_n} carries the test-side state dependence. Under i.i.d. calibration $U_{(k)}$ has the familiar beta law and τ is the identity, recovering ordinary conformal coverage. Under dependent calibration the law of $U_{(k)}$ is generally not beta, and its leading variance is set by the long-run variance of the coverage indicator, computed in Section 6. This is the transported-beta separation (Ramos et al., 2026), to which the Poisson representation adds the explicit corrector and a martingale representation of the calibration term.

5 The first-order coverage expansion

Fix $p = 1 - \alpha$ and $q_p = F^{-1}(p)$, and abbreviate the transient corrector

$$b_{\ell,w}(q) = P^\ell H_q(w) - F(q) = (P^\ell - P^{\ell+1})\chi_q(w). \quad (8)$$

The dependent empirical quantile obeys a Bahadur representation

$$F(R_{(k_p)}) - p = -[F_n(q_p) - p] + o_p(n^{-1/2}), \quad F_n(q) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{R_i \leq q\}, \quad (9)$$

valid under geometric ergodicity with F differentiable at q_p and $f(q_p) > 0$ (Glynn and Henderson, 2002). The expansion below differentiates the corrector in the threshold, so it needs more than (9) supplies. We assume in addition that, in a neighbourhood of q_p , the map $q \mapsto b_{\ell,w}(q)$ is continuously differentiable uniformly in w , that F is continuously differentiable with f bounded away from zero there, and that $R_{(k_p)} \rightarrow q_p$ in probability, which the same geometric ergodicity gives. For a finite chain the first condition reduces to continuous differentiability of each F_w near q_p , since $b_{\ell,w}$ is then a finite linear combination of the F_w . Under these, expanding (6) around q_p and using (9),

$$C_{n,\ell,k_p} - p = b_{\ell,W_n}(q_p) - \left[1 + \frac{1}{f(q_p)} \partial_q b_{\ell,W_n}(q_p)\right] [F_n(q_p) - p] + o_p(n^{-1/2}). \quad (10)$$

When the transient correction is small in C^1 , $\|b_{\ell,\cdot}\|_{C^1} = o(1)$, this reduces to the central formula

$$C_{n,\ell,k_p} - p = b_{\ell,W_n}(q_p) - [F_n(q_p) - p] + o_p(\|b_{\ell,\cdot}\| + n^{-1/2}). \quad (11)$$

Realized coverage error has two first-order sources, a current-state bias $b_{\ell,W_n}(q_p)$ and a calibration-CDF error $F_n(q_p) - p$. The first-order probability-level correction is $p \mapsto p - b_{\ell,W_n}(q_p)$. If the current state causes undercoverage then $b_{\ell,W_n}(q_p) < 0$ and the required empirical quantile level rises. A one-sided honesty margin is then added on top, and its size is set by the same operator.

6 One Poisson equation

No second Poisson equation is required. Center the coverage indicator at the invariant quantile, $g_p(w) = H_{q_p}(w) - p$. Because $F(q_p) = p$, equation (2) at $q = q_p$ reads $(I - P)\chi_{q_p} = H_{q_p} - p = g_p$, so by uniqueness of the centered solution the calibration-variance potential is the corrector itself,

$$\psi_p = \chi_{q_p}. \quad (12)$$

The single solution family $q \mapsto \chi_q$ therefore supplies both objects,

$$b_{\ell,w}(q) = (P^\ell - P^{\ell+1})\chi_q(w), \quad \sigma^2(q) = \pi[\chi_q^2 - (P\chi_q)^2]. \quad (13)$$

The additive calibration error has the exact martingale decomposition, indexed to match the calibration block $i = 1, \dots, n$,

$$\sum_{i=1}^n g_p(W_i) = P\chi_{q_p}(W_0) - P\chi_{q_p}(W_n) + \sum_{i=1}^n D_i, \quad D_i = \chi_{q_p}(W_i) - P\chi_{q_p}(W_{i-1}), \quad (14)$$

where (D_i) are martingale differences for the natural filtration and the endpoint term is bounded by $2\|\chi_{q_p}\|_\infty$. The long-run variance is

$$\sigma_p^2 = \sigma^2(q_p) = p(1-p) + 2 \sum_{j \geq 1} \text{Cov}_\pi(\mathbf{1}\{R_0 \leq q_p\}, \mathbf{1}\{R_j \leq q_p\}). \quad (15)$$

So one operator supplies the transient bias $b_{\ell,w}$ and the calibration variance σ_p^2 . These are not unrelated corrections. The corrector measures how much fast-state information survives to the test horizon. The variance measures how much fast-state persistence removes calibration information. For a reversible chain they are the same coefficients read two ways.

Proposition 2 (Reversible spectral reading). *Let P be reversible with respect to π and suppose its spectrum is pure point, with real eigenvalues $1 = \lambda_1 > \lambda_2 \geq \dots$ and a π -orthonormal eigenbasis ϕ_j of $L_0^2(\pi)$. A finite irreducible aperiodic chain, or any P with compact resolvent, satisfies this. For a chain with continuous spectrum, replace the sums below by integrals against the spectral measure of g_q , which leaves both identities intact in that form. Expand the centered coverage function $g_q = H_q - F(q) = \sum_{j \geq 2} c_j(q) \phi_j$. Then*

$$b_{\ell,w}(q) = \sum_{j \geq 2} c_j(q) \lambda_j^\ell \phi_j(w), \quad \sigma^2(q) = \sum_{j \geq 2} c_j(q)^2 \frac{1 + \lambda_j}{1 - \lambda_j}. \quad (16)$$

Proof. Solving (2) gives $\chi_q = \sum_{j \geq 2} c_j(q)(1 - \lambda_j)^{-1} \phi_j$, so $(P^\ell - P^{\ell+1})\chi_q = \sum_j c_j(q) \lambda_j^\ell (1 - \lambda_j)(1 - \lambda_j)^{-1} \phi_j$, the first identity. For the second, $\pi[\chi_q^2 - (P\chi_q)^2] = \sum_j c_j(q)^2 (1 - \lambda_j^2)(1 - \lambda_j)^{-2} = \sum_j c_j(q)^2 (1 + \lambda_j)/(1 - \lambda_j)$. $\square$

The same coefficients $c_j(q)$ and eigenvalues λ_j control the decay of current-state memory through λ_j^ℓ and the inflation of calibration variance through $(1 + \lambda_j)/(1 - \lambda_j)$. The slogan that state bias and effective-sample-size loss are two readings of one fast mode is exact when a single non-unit eigenvalue dominates, as in the two-state model below, and otherwise they are two functionals of the same spectrum.

7 A fast two-state volatility model

The cleanest one-mode instance is a fast volatility regime. Let $V_t^\varepsilon \in \{L, H\}$ be a continuous-time chain with generator

$$Q_\varepsilon = \frac{1}{\varepsilon} \begin{pmatrix} -a & a \\ b & -b \end{pmatrix}, \quad \kappa = a + b, \quad \pi_H = \frac{a}{a + b}, \quad \pi_L = \frac{b}{a + b}. \quad (17)$$

Conditional on the regime the score CDFs are $F_L(q)$ and $F_H(q)$, and the invariant score law is the mixture $F(q) = \pi_L F_L(q) + \pi_H F_H(q)$. We assume emissions are conditionally independent across time given the regime sequence, or equivalently that any remaining dependence has been folded into the augmented Markov state, so that (21) below holds. If the filtered high-state probability at time zero is $\omega_0 = \mathbb{P}(V_0 = H \mid \mathcal{F}_0)$, then at horizon t

$$\omega_t = \pi_H + e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H), \quad (18)$$

so the exact conditional score CDF and transient coverage error are

$$G_t^\varepsilon(q) = F(q) + e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H)[F_H(q) - F_L(q)], \quad b_t^\varepsilon(q) = G_t^\varepsilon(q) - F(q). \quad (19)$$

This is a literal homogenization boundary layer. After several fast mixing times it is exponentially small. At a forecast horizon comparable to ε it is not small at all. Where F is differentiable with $f > 0$ near q_p , and treating $e^{-\kappa t/\varepsilon}$ as small (that is, after enough fast mixing, not uniformly at $t = 0$), the quantile correction is

$$q_{t,p}^\varepsilon = q_p - \frac{b_t^\varepsilon(q_p)}{f(q_p)} + O(e^{-2\kappa t/\varepsilon}), \quad f = \pi_L f_L + \pi_H f_H. \quad (20)$$

For a nonnegative absolute-error score the high-volatility state usually has $F_H(q) < F_L(q)$ at the relevant q . If $\omega_0 > \pi_H$ then $b_t^\varepsilon(q_p) < 0$ and (20) raises the threshold, as it should.

Discrete sampling and calibration dependence. Sample every Δ time units. The nontrivial eigenvalue of the sampled transition matrix is $\lambda = e^{-\kappa\Delta/\varepsilon}$. Writing $d_p = F_H(q_p) - F_L(q_p)$, the coverage indicator $I_j = \mathbf{1}\{R_j \leq q_p\}$ has

$$\text{Cov}(I_0, I_j) = \pi_L \pi_H d_p^2 \lambda^j, \quad j \geq 1, \quad \sigma_p^2 = p(1-p) + 2\pi_L \pi_H d_p^2 \frac{\lambda}{1-\lambda}. \quad (21)$$

One fast eigenmode produces both effects, the transient bias $\propto \lambda^m(\omega_0 - \pi_H)d_p$ and the variance inflation $\propto \pi_L \pi_H d_p^2 \lambda/(1-\lambda)$, in agreement with Proposition 2 at a single mode. When $\Delta \ll \varepsilon$ we have $1-\lambda \sim \kappa\Delta/\varepsilon$, so $\lambda/(1-\lambda) \sim \varepsilon/(\kappa\Delta)$. Sampling faster piles up redundant calibration scores, and for a window of fixed physical duration $T = n\Delta$ the effective sample size saturates at order T/ε rather than growing like n .

A numerical illustration. Take $p = 0.9$, $\pi_H = 0.2$, $\lambda = 0.8$, and at the invariant 0.9-quantile $F_L(q_p) = 0.95$, $F_H(q_p) = 0.70$, so the mixture is $0.8(0.95) + 0.2(0.70) = 0.90$. If the current state is high volatility the one-step coverage of the invariant threshold is $0.9 + 0.8(1-0.2)(0.70-0.95) = 0.74$. If it is low volatility it is $0.9 + 0.8(0-0.2)(0.70-0.95) = 0.94$. The same pooled threshold gives either 74% or 94% conditional coverage while appearing perfectly calibrated at 90% in stationarity. The long-run variance is $\sigma_p^2 = 0.09 + 2(0.8)(0.2)(0.25)^2(0.8/0.2) = 0.17$, an integrated dependence factor $\sigma_p^2/[p(1-p)] \approx 1.89$, so the nominal calibration sample behaves like only $\approx 53\%$ as many independent observations. A direct simulation of the sampled chain reproduces σ_p^2 (Section 16). The persistent regime creates severe current-state undercoverage and nearly halves the effective calibration sample at once.

8 Homogenization: the continuous-time limit

The two-state calculation is the visible corner of a general fast-slow limit. Let Q be the generator of an ergodic continuous-time chain with invariant law π , sampled every Δ at fast scale ε , so the sampled kernel is

$$P_{\Delta, \varepsilon} = e^{(\Delta/\varepsilon)Q}, \quad \delta = \Delta/\varepsilon. \quad (22)$$

For the centered coverage function $g_q = H_q - F(q)$ define the continuous-time cell problem

$$-Q\varphi_q = g_q, \quad \pi\varphi_q = 0. \quad (23)$$

Proposition 3 (Poisson-to-cell convergence). *On the centered subspace, as $\delta \downarrow 0$,*

$$\delta\chi_q \longrightarrow (-Q)^{-1}g_q = \varphi_q, \quad (24)$$

where χ_q solves (2) for $P_{\Delta, \varepsilon}$. If Q is reversible, the per-observation long-run variance obeys $\delta\sigma_\delta^2(q) \rightarrow 2\langle g_q, (-Q)^{-1}g_q \rangle_\pi$, so for a calibration window of physical duration $T = n\Delta$,

$$\text{Var}(F_n(q)) \approx \frac{2\varepsilon}{T} \langle g_q, (-Q)^{-1}g_q \rangle_\pi. \quad (25)$$

At physical test horizon $t = m\Delta$ the transient corrector is the fast boundary layer $b_t^\varepsilon(q, w) = e^{tQ/\varepsilon}g_q(w)$.

Proof. On the centered subspace $I - e^{\delta Q} = -\delta Q(I + O(\delta))$, so $\delta(I - e^{\delta Q})^{-1} \rightarrow (-Q)^{-1}$ and $\delta\chi_q = \delta(I - e^{\delta Q})^{-1}g_q \rightarrow (-Q)^{-1}g_q$. For reversible Q , $\sigma_\delta^2(q) = \langle g_q, g_q \rangle_\pi + 2\sum_{j \geq 1} \langle g_q, e^{j\delta Q}g_q \rangle_\pi$, a Riemann sum for $\delta^{-1} \int_0^\infty 2\langle g_q, e^{sQ}g_q \rangle_\pi ds = \delta^{-1} \cdot 2\langle g_q, (-Q)^{-1}g_q \rangle_\pi$. Dividing by $n = T/\Delta$ gives (25). The boundary layer is $P_{\Delta, \varepsilon}^m g_q = e^{m\Delta Q/\varepsilon}g_q$. $\square$

Three readings follow. The discrete Poisson solution converges to the cell solution. The honesty variance converges to the Green–Kubo coefficient. The current-state correction is the fast boundary layer. The two-state model of Section 7 is the one-mode corollary, and (25) is the general form of the effective-sample-size saturation at order T/ε . A slow state X_t^ε can be added through a frozen- x generator Q_x and its cell problem, with a remainder controlling the drift of x over the calibration and test horizons. We do not carry that step here.

9 GARCH: the corrector made visible

The two-state model of Section 7 is a cartoon. The reading it illustrates is easiest to see on a process practitioners actually fit. Take GARCH(1,1),

$$r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \omega = 1 - \alpha - \beta,$$

so the stationary variance is one, with score $R_t = |r_t|$ and a pooled split-conformal threshold at level $p = 0.9$ from $n = 2000$ calibration points. GARCH is convenient here because σ_{t+1} is measurable at time t , so the conditional coverage of a threshold T is exactly $2\Phi(T/\sigma_{t+1}) - 1$ with no estimation anywhere. It is also the adversarial case for the averaged reading, because daily equity persistence $\alpha + \beta \approx 0.99$ gives a relaxation half-life near seventy days against a one-day horizon, which is deep inside the boundary layer.

$\alpha + \beta$	half-life	calmest decile	median decile	most volatile decile	marginal
0.800	3.1	0.92	0.91	0.86	0.900
0.900	6.6	0.93	0.91	0.84	0.899
0.960	17.0	0.96	0.91	0.80	0.900
0.990	69.0	0.98	0.92	0.74	0.904
0.997	230.7	0.99	0.93	0.71	0.901
0.999	692.8	0.99	0.93	0.72	0.896

Deciles are of the current volatility state, nominal coverage is 0.90, and the last column is the marginal rate. Read the rows two ways. The marginal guarantee holds exactly everywhere, so conformal does precisely what it promises and no row is evidence of a bug. And the spread across a row is the corrector of Theorem 1: it vanishes as the volatility mixes faster, and at the persistence actually fitted to daily equities the same interval delivers 98% coverage in calm markets and 74% in volatile ones.

That is the whole reading in one table. Pooling averages the fast variable, averaging is valid when the fast variable is fast, and the error is the corrector. The spread grows monotonically with the relaxation time and then saturates once the half-life approaches the calibration window, where the sample no longer sees the stationary law. We swept persistence by moving both GARCH parameters, which also moves the stationary volatility-of-volatility, so the apparent square-root scaling of the spread in the half-life should not be quoted without a design that holds the latter fixed.

10 A finite-sample honest Markov correction

The expansion motivates a finite-sample theorem. We ask for protection at one invariant probability level, not uniformly over every threshold, which is both sharper and cleaner.

Lemma 4 (One-sided Freedman corollary). *Let $(D_i)_{i \leq n}$ be martingale differences with $|D_i| \leq c$ and predictable quadratic variation $\sum_i \mathbb{E}[D_i^2 | \mathcal{F}_{i-1}] \leq nV$. Then for $x = \log(1/\eta)$, with probability at least $1 - \eta$, $\sum_{i=1}^n D_i \leq \sqrt{2nVx} + \frac{c}{3}x$.*

Write $x = \log(1/\eta)$ and, with $B = \|\chi_q\|_\infty$ and V a uniform bound on the conditional increment variance (specified per model below),

$$\delta_{n,\eta} = \frac{2B + \sqrt{2nVx}}{n} + \frac{2Bx}{3n}. \quad (26)$$

For a model class Θ of dependent score dynamics define the target-specific worst-case level

$$u^* = \inf \left\{ u \in [p, 1) : \inf_{\theta \in \Theta} \inf_w P_\theta^\ell H_{\theta, F_\theta^{-1}(u)}(w) \geq p \right\}, \quad (27)$$

the smallest invariant level such that, whatever the model and current state, the ℓ -step coverage of the corresponding invariant quantile is at least p .

Theorem 5 (Finite-sample class-uniform honest correction). *Suppose each F_θ is continuous and strictly increasing on the relevant range, that $u^* + \delta_{n,\eta} < 1$, and that the increments $D_i = \chi_{\theta, F_\theta^{-1}(u^*)}(W_i) - P_\theta \chi_{\theta, F_\theta^{-1}(u^*)}(W_{i-1})$ obey $|D_i| \leq 2B$ and conditional variance at most V , uniformly over $\theta \in \Theta$. Set $k = \lfloor n(u^* + \delta_{n,\eta}) \rfloor + 1$, with $R_{(k)} = +\infty$ if $k > n$. Then*

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta \left\{ \mathbb{P}_\theta(R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n) \geq p \right\} \geq 1 - \eta. \quad (28)$$

The procedure uses only the rank k , not the model.

Proof. Fix θ and write $u = u^*$, $q_u = F_\theta^{-1}(u)$, which is finite because $u < 1$ and F_θ is continuous and strictly increasing, and $F_\theta(q_u) = u$ so $g_u = H_{q_u} - u$ is centered. By (14) and Lemma 4, with probability at least $1 - \eta$,

$$F_n(q_u) \leq u + \delta_{n,\eta}.$$

On this event $nF_n(q_u) \leq n(u + \delta_{n,\eta}) < k$, so fewer than k calibration scores lie at or below q_u , hence $R_{(k)} \geq q_u = F_\theta^{-1}(u^*)$. Since $q \mapsto P_\theta^\ell H_q(w)$ is nondecreasing,

$$\mathbb{P}_\theta(R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n) = P_\theta^\ell H_{R_{(k)}}(W_n) \geq P_\theta^\ell H_{F_\theta^{-1}(u^*)}(W_n) \geq p,$$

the last step by the definition (27) of u^* . The bound is uniform in θ . $\square$

Corollary 6 (Uniform-in-threshold form). *With $a_\ell = \sup_{w,q} [F(q) - P^\ell H_q(w)]_+$ and $u = p + a_\ell < 1$, the choice $u^* \leq p + a_\ell$ recovers the simpler calibration level $p + a_\ell$, since $P^\ell H_{F^{-1}(p+a_\ell)}(w) \geq F(F^{-1}(p+a_\ell)) - a_\ell = p$.*

This is a genuine calibration-sample honest and state-conditionally valid guarantee, stronger than a statement about average marginal coverage, and uniform over the class. One caution. The procedure is honest because k depends only on the fixed class constants u^*, B, V . If instead P, F , and the Poisson solution are estimated from the same data, finite-sample validity is not automatic. It then requires deterministic uniform class bounds, a simultaneous confidence set for the dynamics, or a separate estimation block with a valid envelope.

11 Four attempts that failed

Theorem 5 is correct, but its levels are class constants computed from a known kernel, so it is not a procedure anyone can run. Four attempts to turn the corrector into one are recorded here, with what killed each. The scripts are in the repository and `EXPERIMENTS.md` carries the full numbers.

1. Index the calibration level by the observed state. Keep a pooled calibration block and choose, for each state, the smallest level whose invariant quantile is ℓ -step adequate from that state, paying a union bound over states inside (26). This is honest, by Theorem 5 applied simultaneously at M fixed thresholds. It is also dominated by a simpler scheme that needs no Poisson solution at all: pair each calibration score with the state ℓ steps earlier and take an ordinary quantile within each lagged-state stratum. The indexed scheme can only slide along the quantile function of the *invariant* law, so in a state that already over-covers its level cannot fall below p and the construction stops, while stratifying estimates the ℓ -step law directly and continues. In the two-state model at $\pi_H = 0.2$ the indexed scheme has expected threshold 1.155 against the stratified scheme's 1.002 and over-covers at 0.940 where stratifying is exact at 0.900. No sample size we tried reverses it.

2. Claim the sample-size advantage over stratifying. Stratifying splits the block, so a stratum holds about $n\pi(w)$ points, while a level-indexed pooled scheme uses all n and pays only $\sqrt{\log M}$ in the deviation. This is not a statistical gain. The pooled scheme is handed the per-state levels as population objects computed from a known kernel, so it never pays to learn M quantiles. Minimax lower bounds for conditional conformal coverage over a class of dimension d scale as $\sqrt{d/n}$, and a scheme estimating the same levels would pay $\sqrt{M/n}$ exactly as stratifying does. The apparent saving is the assumption, not the method.

3. Claim that stratifying removes the need for the correction. In the two-state model the independent-data margin survives a pooled variance inflation of 223 once the block is stratified, which suggested the correction was redundant. That is an artefact of the model. A two-state chain has one non-unit mode and it *is* the state, so stratifying removes the dependence by construction. On a continuous state the stratification must bin, and binning is approximate: on an AR(1) log-volatility state at $\phi = 0.98$ with two bins the independent margin fails on 24% of calibration draws against a nominal 5%, and the correction repairs it. The failure decays as the binning refines and is gone by ten bins. The honest statement is conditional: stratifying substitutes for the correction only when it is fine enough to resolve the state.

4. Explain the residual by clustering of stratum visits. Members of a lagged-state stratum arrive at return times whose mean is M but whose median is 1 at high persistence, with over half of consecutive members adjacent in time. Since the autocorrelation across a gap is convex, Jensen gives $\mathbb{E}[\phi^\tau] > \phi^{\mathbb{E}[\tau]}$, which looked like a clean account of why stratifying buys less decorrelation than its sample-size reduction suggests. It is wrong. Kac's lemma forces every equal-probability stratum to the same mean return time, so if visit clustering drove the within-stratum variance then strata with matching gap statistics would have matching variance. At $\phi = 0.98$ with eight bins the mean gap varies by a factor 1.03 across strata and $\mathbb{E}[\phi^\tau]$ by 1.05, while the within-stratum long-run variance of the coverage indicator varies by orders of

magnitude. What that variance tracks is $p_b(1 - p_b)$ at each stratum's own conditional coverage, which runs from 1.0000 in the calmest to 0.4906 in the most volatile, and the ratio of long-run to Bernoulli variance sits near one in six of the eight strata. Almost no serial dependence survives within a stratum. The correct object is the long-run covariance of the coverage indicator, which is classical, and the extremal index is also wrong, being defined only for sets of vanishing measure.

What the four have in common. Each tried to convert an exact description of the corrector into a procedure that beats an existing one, and each was beaten by something simpler or was measuring an artefact of the two-state cartoon. The description survives. The procedures do not.

12 Interpretation

The corrected empirical quantile level is

$$p + \underbrace{a_\ell}_{\text{fast-state protection}} + \underbrace{\delta_{n,\eta}}_{\text{dependent-calibration protection}}, \quad (29)$$

and in threshold units the first-order inflation is $(a_\ell + \delta_{n,\eta})/f(q_p)$. The two costs are qualitatively different. The transient term a_ℓ is not an estimation error. It is the real difference between the current conditional law and the invariant law. The term $\delta_{n,\eta}$ is the price of demanding that the realized calibration sample be trustworthy with probability $1 - \eta$. A state-adaptive implementation need not pay the worst-case a_ℓ . It can estimate $b_{\ell,W_n}(q) = (P^\ell - P^{\ell+1})\chi_q(W_n)$ and build a lower confidence envelope for the current conditional CDF. Theorem 5 is the simplest version with a transparent proof.

13 Combining with the local sharpness modulus

The companion note (Cotton, 2026) shows that honest coverage at a fixed covariate value x_0 forces inflation at the local nonparametric modulus $(ng(x_0))^{-s/(2s+d)}$, from locally indistinguishable alternatives rather than any exchangeability phenomenon. The dependent extension is schematic, and we label it as such. For a kernel K_h the relevant local object is the weighted long-run variance

$$\Omega_h(x_0) = \sum_{j \in \mathbb{Z}} \text{Cov} \left(K_h(X_0 - x_0)[\mathbf{1}\{R_0 \leq q_p(x_0)\} - p], K_h(X_j - x_0)[\mathbf{1}\{R_j \leq q_p(x_0)\} - p] \right), \quad (30)$$

and the argument needs a local Markov mixing CLT of the form $h^d \Omega_h(x_0) \rightarrow \omega_p(x_0)$.

Remark 7 (Schematic bias–margin decomposition). Under a fixed uniform spectral gap the effective local sample stays of order $ng(x_0)h^d$, and the natural upper bound is, in threshold units,

$$T_{n,h}(x_0) - q_p(x_0) \lesssim C_q L h^s + \frac{\sqrt{\omega_p(x_0) \log(1/\eta)}}{f_{p,x_0} \sqrt{ng(x_0)h^d}} + \frac{a_{\ell,x_0}}{f_{p,x_0}} + \frac{R_{\text{hom}}(\varepsilon) + \Delta_{\text{dyn}}}{f_{p,x_0}}, \quad (31)$$

all terms on the threshold scale. Minimizing the first two over h gives $h_\star \asymp [\omega_p(x_0)/(ng(x_0)L^2)]^{1/(2s+d)}$ and local inflation $\asymp L^{d/(2s+d)}[\omega_p(x_0)/(ng(x_0))]^{s/(2s+d)}$.

So under a fixed spectral gap, dependence changes the information constant through the local long-run variance but leaves the Hölder exponent unchanged. If the mixing time itself diverges with n , h , or ε , then $\omega_p(x_0)$ may diverge and the rate can change. Homogenization adds the transient and dependence terms and does not erase the local modulus. The hierarchy is

$$\begin{aligned}
&\text{local heterogeneity} \rightarrow \text{nonparametric modulus}, \\
&\text{calibration persistence} \rightarrow \text{Poisson long-run variance}, \\
&\text{current fast state} \rightarrow \text{Poisson transient corrector}, \\
&\text{approximating the fast-slow model} \rightarrow \text{homogenization remainder}.
\end{aligned} \tag{32}$$

The first term is unavoidable over a broad Hölder class. The second and third can be quantified and corrected under dynamical structure. The fourth must be included for the guarantee to be honest.

14 What homogenization does and does not buy

Homogenization does not defeat the sharpness lower bound. It partitions the conformal error into unavoidable local ambiguity, correctable dynamical bias, and estimable dependence noise. If the fast-slow dynamics live in a finite-dimensional parametric family, honest uncertainty about the dynamical correction shrinks at the parametric rate $n^{-1/2}$ rather than the nonparametric local rate $n^{-s/(2s+d)}$. That gain does not contradict the companion note. It comes from replacing a broad Hölder class by a much smaller structured experiment. If the transition law, the state-specific score distributions, or the scale field are again allowed to vary nonparametrically with x , the local modulus returns, and if their smoothness is unknown the no-adaptation obstruction returns with it.

15 The corresponding skaters transform

The correction is directly usable as a recalibration layer. For an existing distributional forecast F_{t+1}^0 , compute the Gaussianized probability integral transform $Z_t = \Phi^{-1}(F_t^0(Y_t))$, and fit a predictable two-state model to the past Z_t with state-specific CDFs K_L, K_H and filtered high-state probability ω_t . With $\omega_{t+1|t} = \pi_H + \lambda(\omega_t - \pi_H)$ the conditional CDF of the next Gaussianized PIT is $K_{t+1|t}(z) = (1 - \omega_{t+1|t})K_L(z) + \omega_{t+1|t}K_H(z)$, and the dynamically corrected predictive CDF and u -quantile are

$$F_{t+1}(y) = K_{t+1|t}[\Phi^{-1}(F_{t+1}^0(y))], \quad \hat{Q}_{t+1}(u) = (F_{t+1}^0)^{-1}[\Phi(K_{t+1|t}^{-1}(u))]. \tag{33}$$

This is a predictable dynamic PIT recalibration, not by itself a finite-sample conformal procedure. The fence-post map estimates a pooled PIT law, and (33) estimates the conditional PIT law after accounting for the current residual regime. To connect it to Theorem 5 the implementation needs a predictable or separately estimated regime model, an uncertainty envelope covering parameter and filtering error, a probability-level margin, and a treatment of ties or discrete forecast distributions. For an honest upper quantile, let $\mathcal{C}_t$ be a confidence set for the dynamic parameters and add a homogenization or misspecification radius r_t ,

$$z_{t+1,p}^{\text{hon}} = \sup_{\theta \in \mathcal{C}_t} K_{\theta,t+1|t}^{-1}(p + r_t), \quad y_{t+1,p}^{\text{hon}} = (F_{t+1}^0)^{-1}[\Phi(z_{t+1,p}^{\text{hon}})]. \tag{34}$$

Because distributional conformal prediction for Markov processes already composes a transition-CDF PIT (Dai et al., 2026), the novelty here is the Poisson state correction and its honesty envelope, not the conditional PIT composition itself.

Empirical protocol. A comparison should include the base forecast, the marginal fence-post correction, an AR correction, the two-state homogenized correction, and the honest two-state correction. The decisive diagnostics are PIT calibration conditional on the filtered state, coverage conditional on recent Z_t and volatility, the lower-tail probability of realized coverage, proper scores such as log score and CRPS, and the extra interval width the honest envelope demands. Aggregate PIT uniformity alone is insufficient, since the two-state example shows pooled 90% calibration concealing conditional coverage from 74% to 94%.

16 Reproducibility

Two short scripts accompany the note. `verify_decomposition.py` checks, on random finite chains, the corrector identity (6) (maximum error 5.6×10^{-16}), the identity $\psi_p = \chi_{q_p}$ of (12) (exactly zero), the reversible spectral identities of Proposition 2 (error $\sim 10^{-16}$), and the generator limit of Proposition 3 (error decaying linearly in δ). `two_state_illustration.py` reproduces the 74%/94% conditional coverages, the long-run variance 0.17, and the effective sample fraction 0.53, and confirms σ_p^2 by direct simulation of the sampled chain. `garch.py` produces the table of Section 9. `mechanism.py` runs both margins in both schemes across the persistence range, and `jensen_audit.py` is the test that retracted the clustering explanation. `compare_arms.py` produces the threshold comparison of Section 11, and `stress.py` is the attempt to break the independent margin in the stratified scheme, which failed at a variance inflation of 223. `EXPERIMENTS.md` records the full sequence, including a variance bug in an earlier draft: a potential computed on the regime alone drops the emission term and understates V by an order of magnitude at low persistence.

17 Conclusion

Pooling calibration residuals averages a fast variable over its invariant law, which is the averaged limit of homogenization. Split conformal prediction is therefore correct to leading order exactly when the fast variable is fast relative to the forecast horizon, and the gap it leaves is a corrector. For a Markov chain that corrector has an exact Poisson representation, and the same solution family supplies the calibration variance, so the transient state bias and the effective-sample-size loss are two readings of one spectrum.

The GARCH table is the reading in one place. At the persistence fitted to daily equities the marginal guarantee holds exactly and conditional coverage runs from 98% to 74% across volatility deciles. The spread is the corrector, and it vanishes in the regime where the averaged limit is valid.

We were unable to turn any of this into a procedure. Four attempts are recorded in Section 11, and each was dominated by something simpler, rested on an assumption that was doing the work, or measured an artefact of a two-state model. Readers looking for a usable dependence correction for conformal calibration should go to Ramos et al. (2026), which treats calibration-conditional coverage under stationary mixing with the long-run indicator variance, and to the classical effective-sample-size literature beginning with Bayley and Hammersley (1946).

References

- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *Annals of Statistics* 51(2), 816–845.
- Chernozhukov, V., Wüthrich, K., and Zhu, Y. (2021). Distributional conformal prediction. *Proceedings of the National Academy of Sciences* 118(48).
- Cotton, P. (2026). Locally honest coverage forces nonparametric inflation: a sharpness lower bound for prediction sets. Working paper.
- Bayley, G. V. and Hammersley, J. M. (1946). The “effective” number of independent observations in an autocorrelated time series. *Journal of the Royal Statistical Society, Supplement* 8(2), 184–197.
- Dai, Wu, and Politis (2026). Distributional conformal prediction for Markov processes. Preprint.
- Freedman, D. A. (1975). On tail probabilities for martingales. *Annals of Probability* 3(1), 100–118.
- Glynn, P. W. and Henderson, S. G. (2002). Quantile estimation for dependent data. In *Analysis of Simulation Output*.
- Meyn, S. and Tweedie, R. L. (2009). *Markov Chains and Stochastic Stability*, 2nd ed. Cambridge University Press.
- Oliveira, R. I., Orenstein, P., Ramos, T., and Romano, J. V. (2024). Split conformal prediction and non-exchangeable data. *Journal of Machine Learning Research* 25.
- Ramos, T., Graziadei, H., and Cabezas, L. (2026). Conformal prediction via transported beta laws. Preprint.
- Zheng, X. and Proutiere, A. (2024). Conformal prediction under Markovian data. Preprint.